

LeagueEWS

Making early-warning claims testable

Keerthivasan Kannan · PyTorch · temporal models · causal evaluation

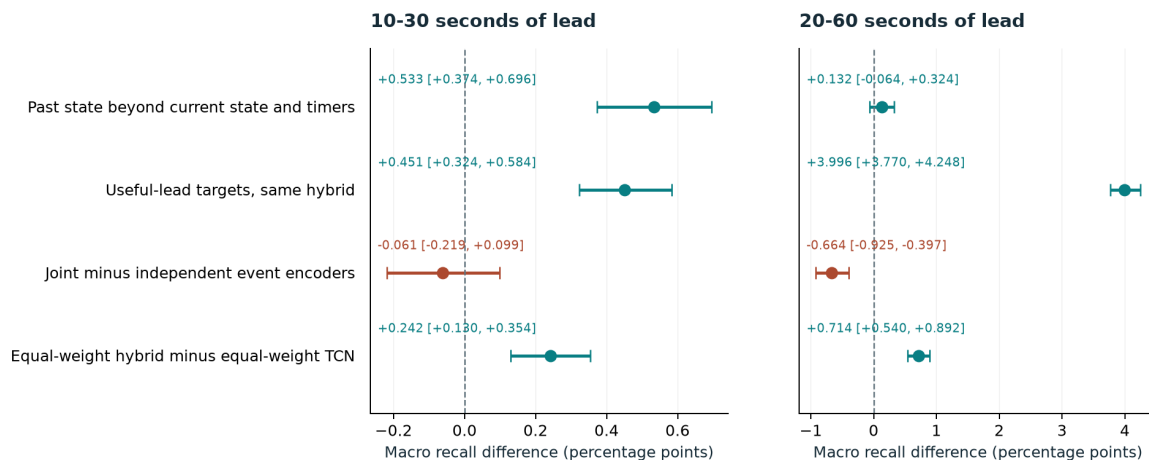
An end-to-end investigation of Baron, Dragon and teamfight warnings in League of Legends: recover a legacy model, audit leakage, rebuild causal inputs, run controlled GPU experiments and measure the cost of useful alerts.

30,000	69	3
development matches	preserved neural fits	event types / fitted seeds

The result

History helps selectively. Shared learning creates task tradeoffs. The hybrid's margin over a strong TCN is small. Meeting an observed alert budget costs recall. The project measures these distinctions rather than relying on the original, contaminated AUC scores.

Which explanations survive controlled comparison?



Exploratory development • conditional intervals • different recipes and contrasts, not additive effects

The figure uses four-budget matched-early operating comparisons. Recall changes are percentage points; the pointwise intervals are conditional on fixed models and policies. All findings are exploratory. No fresh-patch superiority or new-method claim is established.

[Read the paper](#) · [Inspect the exact estimates](#)

From notebook to auditable experiments

Data correctness

Whole-match chronological splits, training-only normalization, native observations and explicit feature availability; objective-completion labels are not misrepresented as engagement onset.

Reliable execution

Checkpointed GPU training, optimizer/RNG resume, source/data hashes, committed scoring releases and complete seed inventories.

Independent verification

The latest policy study records 740 passing tests and 810,000 full scalar replay checks, with exact prior-result reproduction. These are implementation checks, not additional scientific replications.

Model judgment

The equal-weight hybrid uses 1.75M parameters versus TCN's 0.56M. Its +0.242-point short-lead recall margin is reported with regional uncertainty and unequal capacity, instead of being presented as broad superiority.

Measured inference cost

Device / batch	Model	Median across seeds (ms)	Seed median range (ms)
CPU / 1	TCN	0.940	0.733-0.944
CPU / 1	LEAGUEEWS	1.540	1.511-1.674
CUDA / 1	TCN	1.979	1.804-2.050
CUDA / 1	LEAGUEEWS	4.377	4.220-4.567

Six saved fits; resident training inputs; explicit float32; 100 measured calls per cell. Values summarize the three seed-specific medians. CPU: two threads. GPU: RTX 4060 Laptop. Excludes loading, transfers and serving. The full record includes batch 128, p95, all raw timings and numerical checks.

Publication boundary

Working paper, not an accepted publication. The completed evidence supports an exploratory empirical contribution. Fresh confirmation and final literature review remain outstanding; the reserved test is unopened.

AI assistance is disclosed. Project ownership and original notebooks belong to Keerthivasan Kannan; implementation, analysis and writing received material Codex assistance. Human submission review is pending.